

MPS Project Self and Peer Contribution Review (**Confidential**)

Your Name: Tony, Yik-Hau Au

Advisor's Name: David Ruppert

Names of Team Members: Tony Yik-Hau Au (ya293), Wenlin Huang (wh539), Sirui Cao (sc3554), Yuheng Shen (ys2393), Yiyao Jin (yj583)

Project Title: Acreage Estimation and Uncertainty Quantification for New York State

Project Team Number: 2

Today's Date: 12th May 2026

1. Provide an **objective** summary of your own contributions to the whole project. Please complete the following sections. Add space as needed.

A. Data handling:

Downloaded, merged (annual data from 2008-2024 into a master data) and transformed (log-transform, differencing, converted to wide format) the USDA CSB dataset

B. Project design and analysis:

Proposed the idea of the evolution of the modelling (from feature engineering to feature selection from classification to regression model) which served as the flow of the project throughout the semester; Discovered the feature of the data (e.g. spatial correlation, autocorrelation); Built KNN, GARCH, Catboost, LightGBM, XGBoost, CNN, LSTM classification model with evaluation metric (e.g. confusion matrix, ROC & AUC plot, recall & precision & F1 score); proposed idea of exploring conformal prediction, Bayesian model, Kalman filter; Addressed the issue of data engineering (annual CSB mismatch with the monthly weather/soil data) and formatting issue (long format vs wide format)

C. Communications (with client, peers, and advisor):

Arranged meeting agenda; Responsible for non-specific answering question from client and advisor in general;

D. Team meetings:

Leader of the meeting (proposed agenda), provided meeting note for almost all meetings

E. Presentation and poster preparations:

Selected the poster template, finalized the design, responsible for the data exploration, modelling evolution logic, classification model

F. Written report writing:

Provide the plot for my part (classification model and data exploration) and finalized the report; submitted the final delivery

G. Oral presentations:

In charge of Data exploration, feature engineering, classification model, answering technical question (e.g. why don't we use deep learning model; why would tree-based model work in our case)

H. Team leader role:

Arranged group meeting within the group and with the advisor and sponsor; assigned team member role, responsible for non-specific answering question from client and advisor in general, helped groupmate answer question if needed (e.g. interpretation of the model selection and result for the groupmates)

I. Any other comments:

I believe I contributed more than 60% of the group project, largely unwillingly, due to the poor attitude and work quality of the majority of the group. I handled most of the communication within group and with the advisor and client. For example, we established that the rotated team leader should be responsible for the meeting minute and agenda, but the leader at the time didn't follow and thus I have to provide the minute and agenda despite not being the leader, the situation was not improved until the project was completed. I also laid the technical foundation for the study: I data-engineered the large CSB dataset, applied transformations to the data, and produced forecast labels from a classification model for my groupmates' regression model. It became frustrating after a certain point, when some group members stopped responding and lowered their contribution to meetings and project work. I felt that most of the group was relying on me to take the initiative rather than sharing responsibility fairly. I am willing to take on challenges and work on more difficult tasks, so I am fine carrying a heavier share of the project; the main issue is group attitude, in that the majority were unwilling to participate.

2. Provide an **objective** review of the contributions of other team members. Please list them alphabetically by their last names. Add space as needed.

Name of Member 1: Wenlin Huang (wh539)

A. Data handling:

Weather data collection and transformation

B. Project design and analysis:

Proposed linear regression, cubist, xgboost model; but struggled to explain the idea behind the model and the result on why they work or not; lack the statistical modelling skill to choose the right model based on the feature engineering and interpret the result, often rely on advisor's advise. Even still, sometimes he persisted his own model despite the recommendation as if he didn't listen to the advisor and client's advise.

C. Communications (with client, peers, and advisor):

Share his finding with a word document with un-edited R code output (use the crops code instead of the name), seems to read off his script and the words he has put into the script are hard to follow sometimes; Sometimes not respond to the group chat at all despite me asking him specific question; Lack the time-management skills that he has an real person interview which is scheduled months ahead and was in a time-conflict with oral presentation date, but he didn't mention about it until days before the presentation (around 1 week before the oral presentation deadline), created a big struggle to group peers, advisor and client to adjust the meeting plan before the deadline

D. Team meetings:

Only responsible for his own part, barely attempted to contribute for things that are outside of his scope

E. Presentation and poster preparations:

Responsible for weather data exploration but failed to produce his own analysis in the poster, eventually have to use my plot; also responsible for the final nonlinear regression model, slightly inconsistent with my previous feature selection and modeling insight part, due the poor communication within group (he doesn't respond to group message sometimes and barely take responsibility to do more than his part)

F. Written report writing:

Responsible for his own part (weather data and nonlinear model) for plot and result

G. Oral presentations:

Reading off his own script, little to no eye contact; struggled to answer question from advisor and client

H. Team leader role:

Barely take initiative to propose anything, only responsible for organizing meeting during his rotation, barely provide any meeting minute nor agenda

I. Any other comments:

Not the ideal groupmate you want for a project, merely a good follower if you provide the instruction that is under his level of skills, but he is not willing to discuss material beyond his understanding. Not a great presenter as he is just reading off his note with little to no eye contact; struggle to explain and interpret his result and finding; Not a great communicator with poor time management skills and he is non-responsive sometimes; But good for the attendance rate and work ethic

Name of Member 2: Sirui Cao (sc3554)

A. Data handling:

Soil data collection and transformation

B. Project design and analysis:

Proposed multiple linear regression model; but struggled to interpret the result (e.g. why the residual spread, does the model violate the multi-collinearity); lack the statistical modelling skill to choose the model beyond the linear regression; Sometimes rely on advisor's advise. But he is willing to learn and adopt for new material

C. Communications (with client, peers, and advisor):

Share his finding with a word document with un-edited R code output (use the crops code instead of the name); Sometimes not respond to the group chat at all unless there is specific question; willing to charge to organize the meeting

D. Team meetings:

willing to charge to organize the meeting and provide the agenda and meeting minute

E. Presentation and poster preparations:

Responsible for his own part (soil data and linear regression), fair performance

F. Written report writing:

Responsible for his own part (soil data and linear regression), fair performance

G. Oral presentations:

Responsible for his own part (soil data and linear regression), fair performance

H. Team leader role:

Sometimes take initiative to propose something (question, agenda for next meeting), responsible for organizing meeting during his rotation

I. Any other comments:

Fairly good teammate, willing to learn and communicate, able to take initiative, greatly responsive

Name of Member 3: Yuheng Shen (ys2393)

A. Data handling:

Worked on the econ data initially, but then stopped producing any contribution beyond exploration after discovering that data was not suitable for the study in February; barely deal with any technical part of the project

B. Project design and analysis:

Little to no contribution to the model design as he rarely participated in meeting conversion

C. Communications (with client, peers, and advisor):

Barely speak nor ask any questions most of the time; didn't turn on the camera in the first meeting, which was criticized by advisor Dr. Ruppert; He either shows up with nothing to share nor any question to ask, or just not joining the meeting with no reason nor apologies sometimes or skipped the meeting withing days before the meeting with an emergent notice

D. Team meetings:

Barely speak nor ask any questions unless someone asked him directly during the within group meeting

E. Presentation and poster preparations:

Responsible for the objective paragraph and part of the PowerPoint draft, fair; but the weight of his part is significantly less than the rest of the group

F. Written report writing:

Produced the entire report draft, fair performance

G. Oral presentations:

Responsible for his own part (introduction), fair; but the weight of his part is significantly less than the rest of the group

H. Team leader role:

Barely take initiative to propose anything, only responsible for organizing meeting during his rotation, barely provide any meeting minute nor agenda

I. Any other comments:

Barely contribute during meeting and brainstorming; good at following instruction but struggle to take initiative to work beyond his understanding or his part; deadline fighter for his part (e.g. poster, ppt; hours before the actual presentation)

Name of Member 4: Yiyu Jin (yj583)

A. Data handling:

Worked on the VGI data collection but stopped contributing after discovering the data was hard to obtain and engineer in February, barely contributed any technical support beyond that point

B. Project design and analysis:

Little to no contribution to the model design as she rarely participated in meeting conversion

C. Communications (with client, peers, and advisor):

Barely speak nor ask any questions most of the time; didn't turn on the camera in the first meeting, which was criticized by advisor Dr. Ruppert; she either shows up with nothing to share nor any question to ask, or just skipped the meeting withing days before the meeting with an emergent notice

D. Team meetings:

Barely take initiative to propose anything, only responsible for organizing meeting during his rotation, barely provide any meeting minute nor agenda

E. Presentation and poster preparations:

Responsible for the result and summary and part of the PowerPoint draft, fair performance

F. Written report writing:

Responsible for the report draft

G. Oral presentations:

Responsible for her part (summary), fair performance

H. Team leader role:

Barely take initiative to propose anything, only responsible for organizing meeting during his rotation, barely provide any meeting minute nor agenda

I. Any other comments:

Great follower but lack the leadership to take initiative to work beyond her part, barely contribute to brainstorming and meeting; deadline fighter for her part (e.g. poster, ppt; hours before the actual presentation)

Name of Member 5:

A. Data handling:

B. Project design and analysis:

C. Communications (with client, peers, and advisor):

D. Team meetings:

E. Presentation and poster preparations:

F. Written report writing:

G. Oral presentations:

H. Team leader role:

I. Any other comments: